Instructions for homework submission

1. Complete two parts in this homework:
   - Math questions: Include your solution in LaTeX document. Show your work. Submission with embedded photos of handwritten work will not be graded.
   - Programming questions: Complete the given skeleton `Python` code. **For questions requiring visualization, analysis, and discussion, please include your solution in the same LaTeX document.**
2. Submit your work to Gradescope including:
   - A PDF document for written parts: `FirstName_LastName_HW2.pdf`. LaTeX source code is not required.
   - A completed `Python` code: `FirstName_LastName_HW2.py`. *(Note: You may only use standard library (`os`, `pickle`, `re`, etc.), `matplotlib`, and library explicitly defined in given skeleton code.)*
   - **There are two separate submission portals on `Gradescope`:** one for code and one for the report. Submitting your work to the wrong portal will result in a loss of marks.
   - Please assign your answer in PDF report to its corresponding question when submitting to Gradescope. Submitting your work without assigning correspoinding question will result in a loss of marks.
3. This is an individual assignment. While you may consult class mates and other resources, your code should be your own and your write up your own. Please cite any such external help in your write up.
4. Start early!
5. Total: 100 points.

This project utilizes the Bank Marketing dataset[1], provided by UCI Machine Learning Repository. We acknowledge and appreciate their efforts in collecting and maintaining this data.

[1] S. Moro, P. Rita, and P. Cortez. "Bank Marketing," UCI Machine Learning Repository, 2014. [Online]. Available: https://doi.org/10.24432/C5K306.

1 Math Questions

1.1 Problem 1: Information Gain

Note: This is not a programming assignment, so you may **not** use programming tools to solve this problem. Show your work.

Suppose you are given 6 training points as shown below for a classification problem with two binary attributes X_1 and X_2 and three classes $Y \in \{1, 2, 3\}$. You will use a decision tree learner based on information gain.

1. Calculate the information gain for both X_1 and X_2.
2. Report which attribute is used for the first split. Draw the decision tree using this split.
3. Conduct classification for the test example $X_1 = 0$ and $X_2 = 1$.

X_1	X_2	Y
1	1	1
1	1	1
1	1	2
1	0	3
0	0	2
0	0	3

1.2 Problem 2: Entropy

Note: This is not a programming assignment, so you may **not** use programming tools to solve this problem. Show your work.

Suppose you are given the same training points as above for a classification problem with two binary attributes X_1 and X_2 and three classes $Y \in \{1, 2, 3\}$. You will use a decision tree learner based on entropy.

1. Calculate the conditional entropy for both X_1 and X_2.
2. Report which attribute is used for the first split. Draw the decision tree using this split.
3. Conduct classification for the test example $X_1 = 0$ and $X_2 = 1$.

2 Programming Questions

2.1 Part A: Classification Tree (50 points)

In this problem, you will code up a classification tree from scratch. Trees are a special class of graphs with only directed edges and no cycles, also known as directed acyclic graphs (DAGs). Each child node has only one parent node.

1. **Data Processing and EDA:**
   (a) Split the dataset into train and validation sets. Please use a 80:20 split.
   (b) Read and print the training data. Add a short description in your pdf report about the data.
   (c) Extract the features and the label. The label is "`y`".
   (d) Plot histograms of all variables and provide a brief discussion.
   (e) In your data processing class constructor, make call to your defined class methods.
2. **Implementation:**
   (a) Implement a classification tree from scratch. Do not use ML libraries.
   (b) Train the model using training data and validate it with the validation data.
   (c) Aim for `F1 score > 0.4`.

2.2 Part B: Boosting (20 points)

Use a decision-tree-based ensemble algorithm for the `Bank` Dataset.

1. Define a function `train_XGBoost` to train an XGBoost model with L1 regularization. Use bootstrapping (100 iterations) and evaluate different values of α in `alpha_vals = [1e-3, 1e-2, 1e-1, 1, 1e1, 1e2, 1e3]`. Report the optimal α.
2. Initialize `my_best_model` given in the skeleton code with a `XGBClassifier()` on your best hyperparameters. You need to achieve `F1 score > 0.55`.
3. Plot the ROC curve and report the area under the curve (AUC). Include axes labels, legend, and title.

3 Use of AI and External Resources

The purpose of this section is to cite the external resources you used in creating your homework. In particular, if you use any generative AI tools to help generate your code you should then explain:

1. What were the prompts you provided the AI tools?
2. What mistakes did you catch the AI making?
3. What did you do to validate the correctness of the AI generated portions, aside from relying on Gradescope Autograder scoring?